

CONDUCTORS OF ORBIT CLOSURES, AND THE FUNDAMENTAL INVARIANT OF THE 3×3 DETERMINANT

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ABSTRACT. Let v be a form characterised by its symmetries, with finite stabiliser H and affine orbit closure $\overline{\Omega}$. The *boundary deficit* $\text{def}(\lambda, \delta) = \dim S_\lambda^H - \text{mult}_\lambda \mathbb{C}[\overline{\Omega}]_\delta \geq 0$ measures the gap between the stabiliser’s Peter–Weyl count and the coordinate ring of the closure, and the *conductor* $c(\lambda, \delta)$ is the depth of that gap along the boundary, measured in multiplications by a boundary semiinvariant. We compute this invariant exactly in the two smallest power-sum worlds: a closed form in the first, and in the second a transport formula computing the conductor from boundary data alone, on every weight of non-empty support and with no degree bound: the maximal boundary-torus weight divided by the normal weight, corrected by a parity defect supported on one congruence family. The bare floor is not the answer — it first overshoots at $\lambda = (17, 17, 2)$ — and the defect is exactly the obstruction to matching the parity of Young symmetrisation against the residue of the boundary weight. We then compute the conductor for the 3×3 determinant.

Our main results concern $\det_3$. We prove that the minimal degree of a nonzero SL_9 -invariant on $\overline{\Omega} = \overline{\text{GL}_9} \cdot \det_3$ is $e(\det_3) = 18 = 2n^2$, with a one-dimensional space of such invariants, and we evaluate the invariant exactly: it is nonzero at $\det_3$, with value $-2^{16}3^75^37^2$ in a fixed integral normalisation. Bürgisser and Ikenmeyer prove $e(\det_n) \geq n^2$ for every n , strictly when n is odd, and their degree-period theorem confines $e(\det_3)$ to the multiples of 6 that are at least 12; equality $e(\det_n) = n^2$ is known only at $n = 2$ and $n = 4$, in each case by an Alon–Tarsi-type nonvanishing that has no odd analogue. Ours is therefore the third determinant for which the fundamental invariant is determined, and the first whose degree exceeds n^2 ; in particular $e(\det_3) = 18 > 16 = e(\det_4)$, so the invariant degree is not monotone in n . The lower bound $e(\det_3) \geq 18$ turns out to need no computation: an elementary parity count on bracket monomials, which recovers Howe’s vanishing theorems and reproduces the classical first-invariant degrees for ternary cubics and cubic surfaces, already excludes every smaller degree. We show $V(\Phi_{18}) \cap \overline{\Omega} = \partial\overline{\Omega}$ exactly, so that degeneracy of a cubic of border determinantal complexity at most 3 is precisely the vanishing of one explicit degree-18 equation; we deduce that the invariant ring of $\overline{\Omega}$ is a numerical semigroup ring, and, as a demonstration of the method rather than a new result, we recover $\text{per}_3 \notin \overline{\Omega}$ by evaluating a single integer. We then exhibit the second generator of that semigroup ring: the degree period forces an invariant in a degree divisible by 6 but not by 18, the ambient space of degree-24 invariants is one-dimensional — which makes vanishing as cheap to decide as nonvanishing — and $\Phi_{24}(\det_3) = -2^{12}3^75^37^2 \cdot 11 \cdot 41 \neq 0$. Hence $\mathcal{E}(\det_3) \supseteq \langle 18, 24 \rangle$, its complement in $6\mathbb{N}$ is contained in $\{6, 12, 30\}$, and degree 30 is the only value left undetermined. Finally we compute the first conductor of a determinant: $c((2, 2, 2), 2) = 1$, certified at two provably inequivalent orbit points and complete along the entire Φ_{18} -ray. The conductor computation rests on an evaluation functional several of whose natural symmetries we rule out, but whose totals we then determine exactly: they are governed by a gauge Ψ , the restriction of the fundamental degree-6 invariant of $\mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$ to slab normal form. The parabolic character controlling the functional is $\det^2$ in the net’s parameter slot, for the arithmetic reason that the two parts of the relevant weight differ by 2, and a one-dimensional uniqueness statement then forces $\text{TOTAL} = \Psi \cdot 1\,152\,144\,000$.

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1. INTRODUCTION AND SETUP

Let $G = \mathrm{GL}(V)$ act on $W = \mathrm{Sym}^d V^*$, let $v \in W$ have finite stabiliser $H \subset G$, and let $\overline{\Omega} = \overline{G \cdot v}$ be the affine orbit-closure cone. Algebraic Peter–Weyl gives the multiplicities on the orbit,

$$\mathrm{mult}_\lambda \mathbb{C}[G \cdot v] = \dim(S_\lambda^*)^H =: m(\lambda),$$

a purely group-theoretic quantity. The multiplicities of the *closure* are the hard side, and they are the side that carries the obstruction theory of geometric complexity theory [16, 4, 9]. Landsberg’s book [13] is the standard account of the geometry involved, and the programme it describes — separating $\overline{\mathrm{GL} \cdot \mathrm{per}_m}$ from $\overline{\mathrm{GL} \cdot \mathrm{det}_n}$ by representation-theoretic data — is what makes the closure multiplicities worth the trouble; the occurrence obstructions originally proposed for the task are now known not to suffice [4], which sharpens rather than removes the interest in computing the closure side exactly.

Suppose the boundary $\overline{\Omega} \setminus G \cdot v$ is cut set-theoretically by a semiinvariant Δ of degree e and weight det^w . Multiplication by Δ embeds isotypic pieces, so along the ray $(\lambda + mw\mathbf{1}, \delta + me)$ the closure multiplicity is nondecreasing and stabilises at $m(\lambda)$, since $\mathbb{C}[G \cdot v] = \mathbb{C}[\overline{\Omega}][1/\Delta]$.

That the deficit is generally nonzero is not a technicality but the central difficulty of the subject: the orbit closures of det_n and per_n fail to be normal for every $n > 2$ [10], quantitative control on invariants of such closures is delicate [7], and it is precisely this failure that obstructs the Mulmuley–Sohoni programme’s passage from the orbit, where multiplicities are group theory, to the closure, where they are not. The deficit and the conductor are an attempt to measure that failure rather than work around it.

Definition 1.1. The *boundary deficit* is $\mathrm{def}(\lambda, \delta) = m(\lambda) - \mathrm{mult}_\lambda \mathbb{C}[\overline{\Omega}]_\delta \geq 0$, and the *conductor* $c(\lambda, \delta)$ is the least k with $\mathrm{def}(\lambda + kw\mathbf{1}, \delta + ke) = 0$: the maximal pole order along the boundary among functions in the λ -isotypic component.

Remark 1.2 (On the word). The classical conductor is the ideal $\mathfrak{c} = \mathrm{Ann}_O(\tilde{O}/O)$ of a ring in its normalisation, and non-normality of orbit closures is active in this very context, so the choice of name needs justifying. We do not mean $\mathfrak{c}$: here $\mathbb{C}[G \cdot v]$ is a *localisation* of $\mathbb{C}[\overline{\Omega}]$, not its normalisation, and being non-finite over it, the corresponding conductor ideal is zero. What Definition 1.1 measures is the conductor in the numerical-semigroup sense — the least index from which membership holds and continues to hold — and this is literal rather than analogical: Corollary 4.9 shows the invariant ring of $\overline{\Omega}$ is a numerical semigroup ring, and along each ray the deficit vanishes from $c(\lambda, \delta)$ onwards. Equivalently, c is the least k with $\Delta^k \mathbb{C}[G \cdot v]_\lambda \subseteq \mathbb{C}[\overline{\Omega}]$, which is the conductor of $\mathbb{C}[\overline{\Omega}]$ in the module $\mathbb{C}[G \cdot v]$ restricted to one isotypic piece.

That such a k exists, at an explicit uniform bound, is a theorem of Ikenmeyer and Kandasamy [9, Lem. 5.2] for power sums in the regime (number of variables) $\geq$ (degree); their argument is algebraic and makes no optimality claim. Everything in this paper concerns the exact minimal index and the boundary geometry that computes it.

Sections 2 and 3 determine the invariant in the two smallest classical cases, where every ingredient is available in closed form; these serve both as results and as calibration for the determinant. Section 4 contains the main results. Section 5 analyses the evaluation functional used there, including what we could and could not establish about it. Section 7 describes the verification protocol; we consider this part of the content rather than an appendix, since several of the results are computational.

2. BINARY QUARTICS: A CLOSED FORM

Take $W = \text{Sym}^4 \mathbb{C}^2$, $v = x^4 + y^4$, $H = \mu_4^2 \rtimes S_2$ of order 32, so that $\overline{\Omega}$ is the catalecticant cubic hypersurface $\{J = 0\}$. Since $\text{disc} = 256(I^3 - 27J^2)$ identically, disc restricted to $\overline{\Omega}$ equals $256 I^3$: the boundary divisor is cut by the quadratic invariant I (degree 2, weight $\det^4$), along which the discriminant ramifies with index 3, and it is I , not the discriminant, that generates the correct ray. The dense boundary orbit $G \cdot x^3 y$ has stabiliser torus $\text{diag}(t, t^{-3})$; the boundary point is a smooth point of $\overline{\Omega}$ with normal weight of size 8.

Theorem 2.1. *For every weight (a, b) with $a + b = 4\delta$,*

$$\text{def}((a, b), \delta) = c((a, b), \delta) = \max\left(0, \left\lfloor \frac{a-3b}{8} \right\rfloor\right),$$

equivalently $\text{mult}_{(a,b)} \mathbb{C}[\overline{\Omega}]_\delta = N(a, b) - \lfloor (a - 3b)/8 \rfloor^+$, where $N(a, b)$ counts unordered pairs $\{p, q\}$ with $p + q = a - b$ and $p \equiv q \equiv -b \pmod{4}$, excluding $p = q$ when b is odd. Consequently $\text{def} > 0$ if and only if $b \leq \delta - 2$; the maximal conductor at degree δ is $\lfloor \delta/2 \rfloor$; and the total deficit at degree δ is $\lfloor \delta^2/4 \rfloor$.

The proof has two halves. The pole law is computed in the Sylvester model on the transversal family $f_s = x^3 y + s x y^3$, which lies in $\{J = 0\}$ identically with $I(f_s) = -s/4$; the Waring pair is $c = \pm 1/(8\sqrt{s})$, $\ell = x \pm \sqrt{s} y$, and the isotypic component is spanned by the classes $e_{p,q,b} = \text{Sym}[\hat{\ell}_1^p \hat{\ell}_2^q] \cdot (\hat{\ell}_1 \wedge \hat{\ell}_2)^b$. Writing the bracket coefficient at u^k as $A_k(p, q)[\iota^q + (-1)^{b+k} \iota^p]$ with $A_k = [u^k](1+u)^p(1-u)^q$ and $\iota^4 = -1$, the phase survives exactly when $k \equiv b + (p - q)/4 \pmod{2}$, which coincides with the integrality parity; so only $k_0 \in \{0, 1\}$ occurs, and $A_0 = 1$, $A_1 = p - q \neq 0$ away from the identically vanishing class. Hence $\text{pole}(e_{p,q,b}) = \lfloor (p + q - 2b)/8 \rfloor = \lfloor (a - 3b)/8 \rfloor$, uniformly over the classes at a given weight. Second, GL_2 -torus weight spaces are one-dimensional, so all leading terms at a given pole level are proportional to a single monomial and the pole filtration climbs by at most one per level; combined with the identity above, deficit equals conductor equals the pole value. The closed forms follow, the total by $\sum_{b \leq \delta-2} \lfloor (\delta - b)/2 \rfloor = \lfloor \delta^2/4 \rfloor$; this last identity is independently established as an identity of rational functions.

Remark 2.2. The stratified picture closes completely here. The tangent developable's normalisation is $\bigoplus_\delta \text{Sym}^{3\delta} \otimes \text{Sym}^\delta$ under $(\ell, m) \mapsto \ell^3 m$ [1], one copy of each $(4\delta - b, b)$ for $b \leq \delta$; the non-normality gap is exactly one copy of $S_{(4\delta-1,1)}$ per degree — the equivariant globalisation of $k[t^2, t^3] \subset k[t]$ along the cuspidal edge. The full tower reads $\lfloor (a - 3b)/8 \rfloor^+ \rightarrow [b = 1] \rightarrow 0$.

3. TERNARY CUBICS: A TRANSPORT FORMULA

Take $W = \text{Sym}^3 \mathbb{C}^3$, v the Fermat cubic, $H = \mu_3^3 \rtimes S_3$ of order 162, so that $\overline{\Omega}$ is the Aronhold quartic hypersurface $\{S = 0\}$. The space of degree-12 invariants on $\overline{\Omega}$ is one-dimensional, whence $\text{disc}|_{\overline{\Omega}} = c T^2$ for the canonical degree-6 semiinvariant T . The dense boundary orbit is the cuspidal locus $G \cdot (x^2 y + z^3)$: a smooth point of $\overline{\Omega}$ with stabiliser torus $\text{diag}(t, t^{-2}, 1)$, normal weight -6 , and $\nu_\partial(T) = 1$ exactly.

Theorem 3.1. *Write $\mu_{\max}(\lambda) = \lambda_1 - 2\lambda_3$. On every weight with $m(\lambda) > 0$ — with no bound on the degree —*

$$c(\lambda) = \left\lfloor \frac{\mu_{\max}(\lambda)}{|w_N|} \right\rfloor - \pi(\lambda), \quad \pi(\lambda) = \begin{cases} 1, & \lambda_1 = \lambda_2 \text{ and } \mu_{\max}(\lambda) \equiv 1 \pmod{6}, \\ 0, & \text{otherwise,} \end{cases}$$

the maximal boundary-torus weight over the normal weight, corrected by a parity defect. The inequality $c(\lambda) \leq \lfloor \mu_{\max}/|w_N| \rfloor$ holds in general by the block-order estimate below. The hypothesis is stated as $m(\lambda) > 0$ rather than as positivity of the deficit because $m(\lambda)$ is a stabiliser character count — cheap, and computable without any knowledge of $\overline{\Omega}$ — so the theorem can be applied where one does not already know the answer. Remark 3.2 shows the hypothesis cannot simply be dropped.

The defect π is invisible below $\delta = 12$: the smallest weight carrying it with $m(\lambda) > 0$ is $(17, 17, 2)$, where $\mu_{\max} = 13$ and $c = 1$, not 2. On the 254 weights with $m(\lambda) > 0$ and $\delta \leq 10$, π vanishes identically, which is why the uncorrected floor fitted that range exactly.

The proof replaces the pole calculus by a torus computation. The Waring lines of the transversal family $f_s = x^2y + sy^3 + z^3$ are $\hat{\ell}_1 = \kappa\rho^{-1}(x + \rho^3y)$, $\hat{\ell}_2 = -\kappa\rho^{-1}(x - \rho^3y)$, $\hat{\ell}_3 = z$ with $\kappa = 6^{-1/3}$ and $s = \rho^6/3$, so the matrix of Waring lines factors as $A \cdot \text{diag}(\rho^{-1}, \rho^2, 1)$ with A constant: the family is a single τ -orbit of one fixed point $B \cdot v$ of the open orbit. Consequently, for a matrix coefficient $F(g) = \langle \varphi, \pi_\lambda(g)w \rangle$ with w in the multiplicity space S_λ^H ,

$$F(f_s) = \sum_{\nu} \rho^\nu \langle \varphi, \text{pr}_\nu(Bw) \rangle, \quad c(\lambda) = \frac{1}{6} \max\{\nu : \text{pr}_\nu(B \cdot S_\lambda^H) \neq 0\},$$

pr_ν the projection onto the τ -weight- ν subspace. Since the largest τ -weight of S_λ is $\mu_{\max}$, this recovers the upper bound; the content is the identification of the top nonvanishing weight. Realising S_λ^H as the span of the S_3 -symmetrised bracket shapes (a, Q, r) subject to $a_i + \sum_j q_{ij} + r \equiv 0 \pmod{3}$, the slot k left on the untouched line contributes at τ -weights $N_k - 3T$ with $N_k = \mu_{\max} - a_k - 2q_{\bar{k}}$, and two facts close the argument. First, $N_k \equiv 0 \pmod{3}$ for every admissible shape, forced by the μ_3^3 conditions. Second, symmetrisation is a pure parity filter: the two permutations of the k -pair contribute $F(X, Y) + (-1)^{\lambda_1 - a_k} F(-X, -Y)$, so only the drops $T \equiv \lambda_1 - a_k \pmod{2}$ survive. Together these force every achievable ν to be divisible by 6 — the single-valuedness of $F(f_s)$ in s , recovered combinatorially — and reaching $6\lfloor \mu_{\max}/6 \rfloor$ requires $a_k + 2q_{\bar{k}} \in \{\varepsilon, \varepsilon - 3\}$ with $\varepsilon = \mu_{\max} \pmod{6}$. If $\lambda_1 = \lambda_2$ then every $a_k = 0$ and $a_k + 2q_{\bar{k}}$ is even, so for $\varepsilon = 1$ neither value is available and the top is unreachable: this is $\pi(\lambda)$, and it costs exactly one step of the ray. Attainment in every other case is by an explicit shape whose two remaining slots are pushed out of range, so that a single slot contributes and its coefficient is a nonzero multiple of a straightened monomial; the construction needs $\lambda_2 - \lambda_3$ or $\lambda_1 - \lambda_2$ bounded below by an absolute constant, and the finitely many remaining cases are verified directly. That verification is genuinely finite because $m(\lambda)$ and the defect depend only on $(\lambda_1 - \lambda_2, \lambda_2 - \lambda_3, \lambda_3 \pmod{6})$: $\det |_H$ has order exactly 6.

For the upper bound alone one can argue directly on the family: the normalised Waring blocks have exact orders -1 for the two colliding lines, 0 for the untouched line, $+1$ for the collision wedge, -1 for mixed wedges and $+1$ for the determinant. A class with single-counts (a_1, a_2, a_3) , wedge-counts (q_{12}, q_{13}, q_{23}) and determinant power $r = \lambda_3$ therefore has pole $(a_1 + a_2 + q_{13} + q_{23} - q_{12} - r)/6$, whose maximum over class types is $(\lambda_1 - 2\lambda_3)/6$. Since Young symmetrisation only cancels and the first fundamental theorem gives spanning, the bound is rigorous. What that argument cannot see is which cancellations Young symmetrisation actually performs, and the defect π lives exactly there.

Remark 3.2 (Orphans and the failure locus). Attainment of the shadow maximum fails on exactly two kinds of weight: the family $\lambda_1 = \lambda_2$, $\mu_{\max} \equiv 1 \pmod{6}$ of Theorem 3.1, where the parity defect π applies; and weights of empty support, where the λ -isotypic component is zero and there is nothing to attain. The two overlap. The weights with $m(\lambda) = 0$ and shadow pole

≥ 1 are *finitely many* — exactly four, at every degree:

$$(10, 1, 1) [\delta = 4], \quad (13, 1, 1) [\delta = 5], \quad (11, 11, 2) [\delta = 8], \quad (17, 17, 5) [\delta = 13].$$

Finiteness holds because $162 m(\lambda) = \sum_{h \in H} \chi_\lambda(h)$, the three scalars of H contribute $3 \dim S_\lambda$ (as $3 \mid |\lambda|$), the elements with distinct eigenvalues contribute $O(1)$ each by the bialternant, and those with a repeated eigenvalue contribute $O(\lambda_1)$, against a $\dim S_\lambda$ that is cubic in $\lambda_1 - \lambda_2$ and $\lambda_2 - \lambda_3$. The last two orphans are the members of the parity family with $m = 0$; the first two fail instead by cancellation between the three slot classes. They are related: since $\det|_H$ has order exactly 6, $m(\lambda^* \otimes \det^k) = m(\lambda)$ if and only if $6 \mid k$, and that twist exchanges $\lambda_1 - \lambda_2$ with $\lambda_2 - \lambda_3$, so the two $\lambda_1 - \lambda_2 = 0$ orphans are the minimal such twists of the two $\lambda_2 - \lambda_3 = 0$ ones:

$$V_{(10,1,1)}^* \otimes \det^{12} = V_{(11,11,2)}, \quad V_{(13,1,1)}^* \otimes \det^{18} = V_{(17,17,5)},$$

with Weyl dimensions 55, 55 and 91, 91 respectively. *The orphan locus is stable under $\lambda \mapsto \lambda^* \otimes \det^k$ with stabiliser exactly $6\mathbb{Z}$.* Finally, empty support is not the same thing as a shadow that is announced but not delivered: (23, 23, 8), (29, 29, 11) and (35, 35, 14) all have $m(\lambda) > 0$, shadow pole 1, and deficit 0 — they are the members of the parity family whose shadow is 1, sent by Theorem 3.1 to $c(\lambda) = 0$.

Remark 3.3 (What Theorem 3.1 rests on). Three ingredients of the proof are not at the same level of rigour as the rest, and we separate them rather than leave the reader to find out. (i) The dictionary between the indexing of isotypic types used above and the τ -graded model of S_λ was pinned by consistency with the independently proved block-order bound and by the 2361-weight regression, not by tracking the dualisation $V \leftrightarrow V^*$ through the Peter–Weyl identification. This is bookkeeping, but it is bookkeeping we have not done, and it is the one place a sign convention could hide. (ii) Normality of $\overline{\Omega}$ — a hypersurface whose singular locus has codimension three — is used to identify $\mathbb{C}[\overline{\Omega}]$ with the pole-free subring; the codimension statement is standard for $\sigma_3(v_3(\mathbb{P}^2))$ and we quote it. (iii) For the finiteness of the orphan locus we give the character estimate but not the explicit constant that converts it into a bound; the evidence that the list is complete is a sweep to $\lambda_1 - \lambda_2, \lambda_2 - \lambda_3 \leq 150$. None of the three affects the counterexample (17, 17, 2), which refutes the uncorrected equality outright and was confirmed by two independent routes.

Remark 3.4 (Two kinds of tower). The increments along the T -ray are multiplicities of $\mathbb{C}[\overline{\Omega}]/(T)$, the coordinate ring of the cusp cone, bounded by the cusp stabiliser’s invariant counts, with the shortfall being the cusp cone’s own deficit: the structure recurses down the stratification. In the binary-quartic case the recursion terminates after one step. In the ternary-cubic case it does not: the level-two correction admits no floor law and passes to the conic-with-tangent stratum, whose stabiliser is two-dimensional and non-reductive [12]. The deficit there is a lattice count rather than a floor, while the conductor remains a floor — the sharpest structural difference between the two cases. The first case closes level by level only because every parametrisation in its tower is finite.

4. THE DETERMINANT

Let $n = 3$, $V = \mathbb{C}^9$ with coordinates the entries of a 3×3 matrix, and $v = \det_3 \in \text{Sym}^3 V^*$. Write $G = \text{GL}_9$, $\overline{\Omega} = \overline{G} \cdot \det_3$, and let $H \subset G$ be the stabiliser: by Frobenius, $H = ((\text{GL}_3 \times \text{GL}_3)/\mathbb{C}^*) \rtimes \mathbb{Z}/2$ acting by $X \mapsto PXQ$ and $X \mapsto X^t$, of dimension 17. This H is *not* finite, so the deficit of Definition 1.1 is taken with $m(\lambda) = \dim(S_\lambda^*)^H$ as before; the Peter–Weyl side is now a branching count rather than a character count, but the definition and the inequality $\text{def} \geq 0$ are unchanged.

The fundamental invariant. An SL_9 -invariant of degree δ on $\mathrm{Sym}^3 \mathbb{C}^9$ has GL_9 -weight $\det^{\delta/3}$ and corresponds to the rectangular weight $\lambda = ((\delta/3)^9)$; in particular $3 \mid \delta$. Following [3] we call

$$e(v) = \min\{\delta > 0 : \mathbb{C}[\overline{G \cdot v}]_\delta^{\mathrm{SL}_9} \neq 0\}$$

the degree of the fundamental invariant of the orbit closure, and we write $\mathcal{E}(v)$ for the degree monoid $\{\delta : \mathbb{C}[\overline{G \cdot v}]_\delta^{\mathrm{SL}_9} \neq 0\}$.

Three published facts narrow the search before any computation. Howe's theorem [3, Thm. 3.14] gives $\mathbb{C}[\mathrm{Sym}^3 \mathbb{C}^9]_\delta^{\mathrm{SL}_9} = 0$ for $\delta < 9$, and gives 0 again at $\delta = 9$ because the form degree 3 is odd; hence $e(\det_3) > 9$ [3, Thm. 3.15]. Next, $\det_3$ is polystable [3, Cor. 2.9], so the degree monoid generates $b\mathbb{Z}$ for the degree period $b = (m/D)a$ [3, Thm. 3.4], where a is the stabiliser period; since $3 \equiv 3 \pmod{4}$ we have $a(\det_3) = 2$ [3, Thm. 2.5] and therefore

$$b(\det_3) = \frac{9}{3} \cdot 2 = 6, \quad \mathcal{E}(\det_3) \subseteq 6\mathbb{Z}, \quad \gcd \mathcal{E}(\det_3) = 6.$$

So the candidate degrees are $12, 18, 24, \dots$, and the weights $\delta = 3, 9, 15$ carry no invariant for reasons of period alone.

A second, entirely elementary constraint removes the rest of the range. It is a parity count on bracket monomials, and it extends the argument by which Howe disposes of $\delta = m$ when D is odd; we state it in general because the statement costs nothing extra and because it is what does the work here.

Proposition 4.1 (Bracket census). *Let D be odd and let δ, m satisfy $m \mid \delta D$. Put $k = \delta D/m$. If*

$$\delta > \binom{k}{D},$$

then $\mathbb{C}[\mathrm{Sym}^D \mathbb{C}^m]_\delta^{\mathrm{SL}_m} = 0$; consequently $\mathbb{C}[\overline{G \cdot v}]_\delta^{\mathrm{SL}_m} = 0$ for every v .

Proof. By the first fundamental theorem for SL_m [17], the space of invariants of degree δ is spanned by products of $m \times m$ brackets in δ symbolic letters, each letter occurring exactly D times. A bracket is a determinant, so its m letters are distinct; counting incidences, $\delta D = km$ and each letter occupies exactly D of the k brackets. Record this as a 0/1 matrix with rows indexed by letters, row sums D and column sums m .

Suppose two letters $i \neq j$ have equal rows, so that they occur together in the same D brackets and in no others. The transposition exchanging them fixes every bracket of the monomial setwise and swaps two arguments in each of the D brackets containing both, multiplying the monomial by $(-1)^D = -1$. But letters are interchangeable placeholders for one and the same form, so the transposed monomial evaluates to the *same* invariant. That invariant is therefore its own negative, hence zero.

The rows are D -subsets of $[k]$, of which there are $\binom{k}{D}$. If $\delta > \binom{k}{D}$ two rows coincide for every monomial, so every spanning monomial evaluates to zero. The statement for an orbit closure follows because SL_m is reductive, so restriction $\mathbb{C}[\mathrm{Sym}^D \mathbb{C}^m]_\delta^{\mathrm{SL}_m} \rightarrow \mathbb{C}[\overline{G \cdot v}]_\delta^{\mathrm{SL}_m}$ is surjective. $\square$

Two checks, both against classical answers. For ternary cubics ($m = D = 3$) the bound admits $\delta = 4$ and excludes $\delta \leq 3$: the first invariant is Aronhold's S , in degree 4. For cubic surfaces ($m = 4, D = 3$) it admits $\delta = 8$ and excludes $\delta = 4$: the invariant ring of the cubic surface is generated in degrees $8, 16, 24, 32, 40, 100$. The bound also recovers Howe's two statements — for $\delta < m$ no bracket monomial exists at all, and for $\delta = m$ with D odd every letter lies in all $k = D$ brackets, so all rows coincide.

For $\det_3$ we have $m = 9$, $D = 3$, $k = \delta/3$, and the condition $\delta \leq \binom{\delta/3}{3}$ first holds at $\delta = 18$:

δ	3	6	9	12	15	18
$k = \delta/3$	1	2	3	4	5	6
$\binom{k}{3}$	0	0	1	4	10	20
$\delta \leq \binom{k}{3}$	no	no	no	no	no	yes

Corollary 4.2. $e(\det_3) \geq 18$, with no computation.

At $\delta = 18$ the census is silent, and one exact count settles what it leaves open.

Theorem 4.3 (Ambient census). $\dim_{\mathbb{C}} \mathbb{C}[\mathrm{Sym}^3 \mathbb{C}^9]_{18}^{\mathrm{SL}_9} = 1$.

Proof. The dimension is the plethysm coefficient $\langle h_{18}[h_3], s_{(6^9)} \rangle$, evaluated by expanding $h_{18}[h_3]$ in the power-sum basis and pairing against the character $\chi^{(9^6)}$ by the Murnaghan–Nakayama rule. The expansion has 44 237 distinct classes and the value is 1. The computation is exact rational arithmetic throughout and is independent of everything in Sections 5 and 7. $\square$

Definition 4.4. Φ_{18} denotes a nonzero element of $\mathbb{C}[\mathrm{Sym}^3 \mathbb{C}^9]_{18}^{\mathrm{SL}_9}$. By Theorem 4.3 it is determined up to a scalar, and every statement we make about it below — non-vanishing, the divisor, the semigroup, and all ratios of values — is independent of that scalar. Where a bare numerical value appears, the normalisation in force is stated with it.

Three objects in this circle of ideas carry the letter Φ or ϕ , and it is worth separating them. Bürgisser and Ikenmeyer write Φ_w for the fundamental invariant of a polystable form w , of degree $e(w)$ and normalised by $\Phi_w(w) = 1$; their subscript names the *form*, ours names the *degree*, and the two agree up to scale, $\Phi_{18} = \Phi_{18}(\det_3) \cdot \Phi_{\det_3}$. They also write ϕ_w for the invariant of degree $b(w)$ defined on the *orbit* alone by $g \cdot w \mapsto \det(g)^{a(w)}$ [3, Lem. 3.2]; here $b = 6$, and since $\mathbb{C}[\overline{\Omega}]_{18}^{\mathrm{SL}_9}$ is one-dimensional,

$$\Phi_{18}|_{G \cdot \det_3} = c\phi^3 \quad \text{for some } c \in \mathbb{C}^\times.$$

The degree monoid of Corollary 4.9 is exactly $\{6k : \phi^k \text{ extends regularly to } \overline{\Omega}\}$, so a question about which degrees carry invariants is a question about which powers of ϕ extend across the boundary.

This is what makes Φ_{18} an object rather than an output. It is defined by a dimension count, not by the algorithm that evaluates it, and its uniqueness has three consequences we use. First, $e(\det_3) \leq 18$ needs only that Φ_{18} does not vanish identically on $\overline{\Omega}$, and a single nonzero value at a single point of the orbit certifies that. Second, the restriction map $\mathbb{C}[\mathrm{Sym}^3 \mathbb{C}^9]_{18}^{\mathrm{SL}_9} \rightarrow \mathbb{C}[\overline{\Omega}]_{18}^{\mathrm{SL}_9}$ is either zero or injective, so the census reduces $e(\det_3)$ to *one bit*, and one evaluation decides it. Third, since the ambient space is one-dimensional, the value of Φ_{18} at any point of $\mathrm{Sym}^3 \mathbb{C}^9$ — including points outside $\overline{\Omega}$ — is canonical up to the single global scale, which Corollary 4.13 needs.

Theorem 4.5. $\dim_{\mathbb{C}} \mathbb{C}[\overline{\Omega}]_{18}^{\mathrm{SL}_9} = 1$, and therefore

$$e(\det)_3 = 18 = 2n^2.$$

Proof. $\geq$ is Corollary 4.2. For $\leq$, Theorem 4.3 and Definition 4.4 make Φ_{18} the unique candidate up to scale, and $\Phi_{18}(\det_3) \neq 0$ by (1); since $\det_3 \in \overline{\Omega}$, the restriction of Φ_{18} to $\overline{\Omega}$ is not the zero function. One-dimensionality of the restricted space follows from that of the ambient space. $\square$

We are explicit about the dependencies, because the reader who has looked ahead to Section 5 will want to know how much of it Theorem 4.5 rests on. The answer is: none. The lower bound is combinatorial; the ambient count is a plethysm; the upper bound is one integer. The gauge law of Section 5, which is verified and not proved, and the covariance hypotheses refuted there, enter nowhere in this section. Section 5 studies *how* the evaluation behaves across orbit points; whether the invariant exists does not depend on it.

An earlier form of this argument established the vanishing at $\delta = 3, 6, 9, 12, 15$ by direct multiplicity computation, and those runs were in fact carried out; the bracket census makes all five superfluous. We have kept them, and report that they return zero at every one of the five, because a computation that agrees with a proof it did not use is the cheapest calibration available and this one exercises the whole pipeline at the exact weights that matter. The vanishing statements are exact multiplicity computations in $\mathbb{C}[\overline{\Omega}]_\delta$ at the rectangular weights, run in the streamed level algorithm of Section 7; the nonvanishing at $\delta = 18$ is witnessed by the value (1), which is a single exact integer and therefore settles the question on its own.

Theorem 4.6. $\Phi_{18}(\det_3) \neq 0$. *In the integral normalisation $\mathcal{N}$ — the evaluation functional of Section 5 taken with unit weight on the standard slab basis —*

$$\Phi_{18}(\det) = -877\,879\,296\,000 = -2^{16} 3^7 5^3 7^2. \quad (1)$$

Remark 4.7 (What depends on the scale, and what does not). Only the non-vanishing is used in the proof of Theorem 4.5; the integer itself is a statement about $\mathcal{N}$. Under Bürgisser and Ikenmeyer’s convention $\Phi_{\det_3}(\det_3) = 1$ holds by fiat and carries no information, so the scale-invariant content of the two values computed here is their ratio

$$\frac{\Phi_{18}(\text{per}_3)}{\Phi_{18}(\det_3)} = -\frac{272}{4725},$$

which is independent of every choice. The canonical integral choice would be the primitive generator of the rank-one lattice of integral invariants in this degree, unique up to sign — the same convention under which $\Psi = -I_6^{\text{prim}}$ in Section 5. We have not determined whether $\mathcal{N}$ is primitive. The content divides the gcd of the two values, namely $2^{16} 3^4 \cdot 5 \cdot 7$, which is a weak bound; if $\mathcal{N}$ is a proper multiple of the primitive generator then (1) must be divided by that multiple and the factorisation restated. We flag this because the factorisation, rather than the non-vanishing, is the part a reader is likely to quote.

Boundary geometry. Two facts about this boundary are already known, and we use both. Hüttenhain and Lairez [6], answering a question of Landsberg, determine the boundary of the orbit of $\det_3$ completely (see also [8]): it is *pure* of codimension one with exactly two irreducible components, the orbit closures of the determinant of the generic traceless matrix and of the universal quadric $x_4x_1^2 + x_5x_2^2 + x_6x_3^2 + x_7x_1x_2 + x_8x_2x_3 + x_9x_1x_3$. We write P_1 and P_2 for these two components (in [6] the same symbols denote the two forms). Independently, Bürgisser and Ikenmeyer show for any polystable w that the zero set of the fundamental invariant in the orbit closure is exactly the boundary, and that $\mathbb{C}[G \cdot w]$ is the localisation of $\mathbb{C}[G \cdot w]$ at it [3, Prop. 3.9]. Applied to Φ_{18} this gives $V(\Phi_{18}) \cap \overline{\Omega} = \partial\overline{\Omega}$ and puts Φ_{18} in the role of the semiinvariant Δ of Section 1, of degree $e = 18$ and weight $\det^6$. We verified the vanishing directly at both of Hüttenhain and Lairez’s representatives, which is a check on our normalisation rather than a new result.

What is left to compute is how Φ_{18} vanishes, and that is not formal.

Proposition 4.8. *As a divisor on $\overline{\Omega}$,*

$$\text{div}(\Phi_{18}|_{\overline{\Omega}}) = 6P_1 + 9P_2.$$

The two multiplicities arise differently, and the asymmetry is the point. Along P_1 the orbit closure is smooth at a generic point, a transversal one-parameter family meets the divisor transversally, and the order is read off from the weight: an SL_9 -invariant of degree 18 satisfies $\Phi_{18}(g \cdot v) = \det(g)^6 \Phi_{18}(v)$, giving ramification 6. Along P_2 that argument is unavailable — the orbit closure of $\det_n$ is not normal for $n > 2$ [10], and P_2 is where the non-normality sits — and the multiplicity 9 is obtained instead by a computation, from integrality of the torus weights on the component. We record it as a computed value rather than a proved one; deriving it directly from the explicit degeneration to P_2 that Hüttenhain and Lairez supply — via Jacobi’s formula, with A skew-symmetric and S symmetric — is the obvious next step and we have not carried it out. In particular the multiplicities are not both determined by the character, and a reader who expects 6 on both components should note that this is exactly what fails.

Corollary 4.9. *Restriction to the dense orbit is injective on $\mathbb{C}[\overline{\Omega}]^{\mathrm{SL}_9}$ and $\dim \mathbb{C}[\overline{\Omega}]_{\delta}^{\mathrm{SL}_9} \leq 1$ for every δ , so $\mathbb{C}[\overline{\Omega}]^{\mathrm{SL}_9}$ is the semigroup ring of the numerical semigroup $S = \mathcal{E}(\det_3)$. Combining Theorem 4.5 with the degree period, $S \subseteq 6\mathbb{N}$, $\min S = 18$ and $\gcd S = 6$; hence $S \neq 18\mathbb{N}$. By Theorem 4.10, $24 \in S$, and therefore*

$$S \supseteq \langle 18, 24 \rangle = 6 \cdot \langle 3, 4 \rangle = \{0, 18, 24, 36, 42, 48, 54, \dots\}.$$

Equivalently $\frac{1}{6}S$ contains every nonnegative integer except possibly 1, 2, 5, and $1, 2 \notin \frac{1}{6}S$. The only undetermined degree is 30: either $S = \langle 18, 24 \rangle$, or $S = 6 \cdot \langle 3, 4, 5 \rangle = \{0, 18, 24, 30, 36, \dots\}$.

In the first version of this paper the last line of Corollary 4.9 was a pure existence statement: were $S = 18\mathbb{N}$ its greatest common divisor would be 18, while the degree period forces 6, so $\overline{\Omega}$ had to carry an invariant in some degree that is a multiple of 6 but not of 18, and 24 was the smallest candidate. That invariant now exists explicitly.

Theorem 4.10. $\dim_{\mathbb{C}} \mathbb{C}[\mathrm{Sym}^3 \mathbb{C}^9]_{24}^{\mathrm{SL}_9} = 1$; write Φ_{24} for a generator. Its value on the determinant is nonzero:

$$\Phi_{24}(\det) = -24\,745\,222\,656\,000 = -2^{12} 3^7 5^3 7^2 \cdot 11 \cdot 41 \neq 0, \quad (2)$$

in the normalisation fixed by the explicit bracket monomial recorded in `results/results_deg24.md`. Consequently

$$24 \in \mathcal{E}(\det_3), \quad \mathrm{def}((8^9), 24) = 0, \quad \mathrm{mult}_{(8^9)} \mathbb{C}[\overline{\Omega}]_{24} = 1,$$

and Φ_{24} is the second generator of $\mathbb{C}[\overline{\Omega}]^{\mathrm{SL}_9}$ — a new generator, not a product, since there is no invariant of degree 6 on $\overline{\Omega}$ to multiply Φ_{18} by.

The one-dimensionality in Theorem 4.10 is what makes the statement cheap, and it is worth isolating why. For a general degree the vanishing direction is the expensive one: it needs the multiplicity of the rectangular weight in $\mathbb{C}[\overline{\Omega}]_{\delta}$, a computation strictly larger than the $\delta = 18$ census. Here it is free. Since $\mathbb{C}[\overline{\Omega}]_{24}^{\mathrm{SL}_9}$ is the image of $\mathbb{C}[\mathrm{Sym}^3 \mathbb{C}^9]_{24}^{\mathrm{SL}_9}$ under restriction — surjective on invariants because SL_9 is reductive — and that ambient space is a single line, $24 \in \mathcal{E}(\det_3)$ holds *if and only if* $\Phi_{24}(\det_3) \neq 0$. One integer decides both directions. The ambient Hilbert function, computed exactly, is

$$\dim \mathbb{C}[\mathrm{Sym}^3 \mathbb{C}^9]_{\delta}^{\mathrm{SL}_9} = 0, 0, 0, 0, 0, \mathbf{1}, 0, \mathbf{1}, 1, 4 \quad (\delta = 3, 6, \dots, 30).$$

Two features deserve comment. The bracket census of Proposition 4.1 permits $\delta = 21$ — $21 \leq \binom{7}{3} = 35$ — yet the multiplicity there is 0, so the exact count is strictly sharper than the pigeonhole bound. And at $\delta = 30$ the ambient space is four-dimensional, so the same collapse does not occur: degree 30 is not decided by a single number, and Corollary 4.9 leaves it open for that reason.

Remark 4.11 (A normalisation-free form of (2)). The scale of Φ_{24} is a convention, so (2) should be read alongside a ratio that is not. Restrict Φ_δ to the six-dimensional space U of cubics $\sum_{\sigma \in S_3} u_\sigma x_{1\sigma 1} x_{2\sigma 2} x_{3\sigma 3}$, which contains both $\det_3$ and per_3 . The diagonal torus of GL_9 preserves U and the weight condition $\sum_{\sigma} e_{\sigma} M_{\sigma} = 8J$ has exactly the nine solutions $e = (a, a, a, 8-a, 8-a, 8-a)$, so $\Phi_{24}|_U = \sum_{a=0}^8 K_a P^a Q^{8-a}$ with $P = u_{\text{id}} u_c u_{c^2}$ and $Q = u_{t_1} u_{t_2} u_{t_3}$; a row/column permutation of odd total sign exchanges P and Q and acts trivially on $\det^8$, so $K_a = K_{8-a}$. Then $\Phi_{24}(\det_3) = \sum_a (-1)^a K_a$ and K_8 is the value of the run in which only even permutations are allowed, and

$$\Phi_{24}(\det_3)/K_8 = 17325 = 3^2 5^2 7 \cdot 11,$$

independently of the bracket monomial used. The same construction at $\delta = 18$ gives $\Phi_{18}(\det_3)/K'_6 = 4725 = 3^3 5^2 7$ and $\Phi_{18}(\text{per}_3)/K'_6 = -272 = -2^4 \cdot 17$, whose quotient is the ratio $-4725/272$ of (1) and (3). Note also that $K_a = K_{8-a}$ pairs indices of *equal* parity, so it does not annihilate the alternating sum: no symmetry of U forces (2) to vanish. This was recorded as a pre-registered prediction before the value was computed.

Remark 4.12 (Why the divisor does not decide it). It is tempting to settle degree 24 from Proposition 4.8 without any computation. On the orbit there is an invariant φ of degree 6 with $\varphi(g \cdot \det_3) = \det(g)^2$, well defined because $\det(h) = \pm 1$ for $h \in H$, and $\Phi_{18}|_{G \cdot \det_3} = \Phi_{18}(\det_3) \varphi^3$; so $\text{div}(\varphi) = 2P_1 + 3P_2$, and φ^4 has effective divisor. The inference “effective divisor, hence regular” is Serre’s criterion and needs $\overline{\Omega}$ normal, which it is not; applied to φ itself the same inference would give $6 \in \mathcal{E}(\det_3)$, which is false. What the computation does prove is the positive part: φ is regular on the normalisation $\overline{\Omega}'$, so $\mathbb{C}[\overline{\Omega}']^{\text{SL}_9} = \mathbb{C}[\varphi]$ has degree monoid exactly $6\mathbb{N}$, and $\mathbb{C}[\overline{\Omega}]^{\text{SL}_9}$ is the semigroup ring of $S \subseteq 6\mathbb{N}$ inside it. The gaps of $\frac{1}{6}S$ — now known to be $\{1, 2\}$ or $\{1, 2, 5\}$ — are precisely the degrees at which the non-normality of $\overline{\Omega}$ is arithmetically visible on invariants, and the conductor of that numerical semigroup is the conductor of $\mathbb{C}[\overline{\Omega}]^{\text{SL}_9}$ in its normalisation.

Corollary 4.13. $\text{per}_3 \notin \overline{\Omega}$; that is, the border determinantal complexity of the 3×3 permanent exceeds 3.

Proof. By Theorem 4.3 the ambient space of degree-18 invariants is one-dimensional, so Φ_{18} is defined on all of $\text{Sym}^3 \mathbb{C}^9$ and its value at any point is canonical once the scale is fixed; no choice of lift is involved. Evaluating at the permanent gives

$$\Phi_{18}(\text{per}_3) = +50\,536\,120\,320 = 2^{20} 3^4 5 \cdot 7 \cdot 17 \neq 0. \quad (3)$$

By Proposition 4.8, Φ_{18} vanishes identically on $\partial \overline{\Omega}$, so $\text{per}_3 \notin \partial \overline{\Omega}$. If per_3 lay in $\overline{\Omega}$ it would therefore lie in the dense orbit $G \cdot \det_3$, contradicting the theorem of Marcus and Minc [15] that no element of GL_9 carries $\det_3$ to per_3 . Hence $\text{per}_3 \notin \overline{\Omega}$. $\square$

The conclusion of Corollary 4.13 is of course classical and easy by other means; what is new is the route. The obstruction is a single integer, computed once, and the argument uses no asymptotics, no representation-theoretic multiplicity comparison, and no degeneration analysis — exactly the shape of separation that geometric complexity theory asks for — and the shape in which the strongest conditional lower bounds are stated [14] — at the smallest scale where the shape can be exhibited at all.

Remark 4.14 (Comparison). The determinant is the case where the general theory stops short. For even n , $e(\det_n) = n^2$ holds if and only if a signed count of admissible n -tables is nonzero [3, Prop. 3.28], an Alon–Tarsi-type criterion verified at $n = 2$ and, by computer, at $n = 4$; the analogous statement for the product of variables is Kumar’s [10, 11]. For *odd* n the

criterion has no analogue — the degree- n^2 invariant on the ambient space vanishes identically for parity reasons — so nothing in the literature predicts a value, and $e(\det_n)$ was known for no odd n . Theorem 4.5 supplies the first one. Whether $18 = 2n^2$ reflects a pattern or is an accident of $n = 3$ we do not know; the only other exact values are $e(\det_2) = 4$ and $e(\det_4) = 16$, both equal to n^2 , and three points are not a sequence. See Section 8.

The first conductor of a determinant. With Φ_{18} as boundary semiinvariant, Definition 1.1 applies verbatim: the conductor ray attached to (λ, δ) is $(\lambda + 6k\mathbf{1}_9, \delta + 18k)$.

The two halves of the next statement have very different characters, and we separate them because conflating them misrepresents what rests on what.

Proposition 4.15 (The deficit). $\text{def}((2, 2, 2), 2) = 1$.

Proof. Elementary in both factors. For the closure multiplicity, $\text{mult}_\lambda \mathbb{C}[\overline{\Omega}]_2 \leq \text{mult}_\lambda \text{Sym}^2(\text{Sym}^3 \mathbb{C}^9)$, and the classical plethysm $\text{Sym}^2(\text{Sym}^3) = S_{(6)} \oplus S_{(4,2)}$ contains no $S_{(2,2,2)}$ whatever; hence $\text{mult}_\lambda \mathbb{C}[\overline{\Omega}]_2 = 0$ and $\text{def} = m(\lambda)$. For $m(\lambda)$: the identity component of H acts through $\mathbb{C}^9 = \mathbb{C}^3 \otimes \mathbb{C}^3$, so an invariant in $S_{(2,2,2)}(V \otimes W)$ must lie in $S_\mu V \boxtimes S_\nu W$ with both factors one-dimensional, forcing $\mu = \nu = (2, 2, 2)$; the multiplicity is the Kronecker coefficient $g((2, 2, 2), (2, 2, 2), (2, 2, 2)) = 1$. The residual $\mathbb{Z}/2$ generated by transposition acts on that line by a sign, which is $+1$. Hence $m(\lambda) = 1$. $\square$

No orbit point, no evaluation functional, and no computation beyond a Kronecker coefficient enters that proof. The conductor is a different matter.

Theorem 4.16 (The conductor). $c((2, 2, 2), 2) = 1$, and the vanishing is ray-complete: $\text{def}(\lambda + 6k\mathbf{1}_9, 2 + 18k) = 0$ for every $k \geq 1$.

Here the certification is genuinely needed, and here is what it certifies. The $k = 1$ rung asks whether $\text{mult}_\lambda \mathbb{C}[\overline{\Omega}]_{20}$ attains $m(\lambda)$, and the witness is the non-vanishing of one explicit function: the $\lambda' = (8, 8, 8, 6^6)$ highest weight vector in degree $\delta = 20$, whose value is the TOTAL of Section 5. It was evaluated independently at two points of $\overline{\Omega}$, called C and R , which Lemma 5.1 shows lie in distinct H -double cosets; both returned $1\,152\,144\,000 \neq 0$. A point-evaluation cannot certify a deficit, which is a difference of dimensions; what it certifies is that a specific function is not identically zero, hence that a multiplicity is at least one. Ray-completeness for $k \geq 2$ then follows from the stabilisation argument of Section 1 together with the $k = 1$ result.

This is the first conductor of a determinant to be computed exactly, and it is the smallest possible nonzero answer: one multiplication by the fundamental invariant clears the gap, and it stays cleared for ever after. The scales are worth noting — the deficit is detected at degree 2, far below the degree 18 at which the invariant that repairs it first exists.

5. THE EVALUATION FUNCTIONAL

The computations of Section 4 all pass through one object: a $\mathbb{Z}$ -valued functional on the orbit, evaluated at a point u presented in *slab normal form*. Writing a point of $\mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$ as a triple of 3×3 slabs and normalising the first slab to the identity, a normal form is a pair $N = (A, B)$ of 3×3 matrices, and the residual symmetry is simultaneous conjugation $(A, B) \mapsto (gAg^{-1}, gBg^{-1})$ together with the finite symmetries of the slab frame.

Lemma 5.1 (Double cosets). *Write points of the orbit as $g \cdot \det_3$ with $g \in G$. The value of the functional at $g \cdot \det_3$ depends only on the double coset $HgH \in H \backslash G / H$. Moreover the intersection algebra*

$$\mathfrak{s}(g) = \mathfrak{h} \cap \text{Ad}(g)\mathfrak{h} \subseteq \mathfrak{gl}_9, \quad \mathfrak{h} = \text{Lie } H,$$

is carried to $\text{Ad}(h)\mathfrak{s}(g)$ under $g \mapsto hgh'$ for $h, h' \in H$, so its isomorphism type is an invariant of the double coset. Consequently two points whose intersection algebras are non-isomorphic lie in different double cosets, and a value certified at one is an independent certification when reproduced at the other.

We use this in exactly one place, and it is the place that matters. At the two certification points of Theorem 4.16,

$$\dim \mathfrak{s}(g_C) = \dim \mathfrak{s}(g_R) = 4,$$

but $\mathfrak{s}(g_C)$ is non-abelian while $\mathfrak{s}(g_R)$ is abelian. The two points are therefore genuinely inequivalent, and the agreement

$$\text{TOTAL}(g_C) = \text{TOTAL}(g_R) = 1\,152\,144\,000$$

is a check with content rather than a repetition of one computation in two coordinate systems. This is the strongest form of internal verification available to us: no symmetry of the setup carries one point to the other, so the shared value is a statement about the functional and not about the frame. Theorem 5.5 now explains the agreement — the two points have the same value of the gauge — but it does not weaken the check, since the theorem’s normalisation is fixed at C and the value at R was measured before any of it existed.

Proposition 5.2 (Taxonomy of vanishing). *Every vanishing subvalue in the census of Section 4 is accounted for by exactly one of three mechanisms, and the three are mutually exclusive on the cases that occur:*

- (i) mass infeasibility — *no admissible assignment of level masses exists and the sum is empty. This tier is decided by the displacement-cancellation lemma: content is feasible if and only if there exist 2+2 conversions with $\sum(e_{\text{tgt}} - e_{\text{src}}) = 0$, which follows from the forced balanced distribution $t = (2, 2)$ together with an integral Birkhoff decomposition for 3×3 doubly stochastic matrices. It is a proof, not a test, and it derives the observed 15-of-81 sweep structurally;*
- (ii) wedge structure — *assignments exist but every one of them carries a repeated factor in an antisymmetrised slot, so each term vanishes individually;*
- (iii) sign cancellation — *terms exist and are individually nonzero, and the sum vanishes.*

Only (iii) requires the computation; (i) and (ii) are decided by inspection of the mask, and together they account for the great majority of the census.

The taxonomy is what makes the census tractable: it converts most of the vanishing into combinatorics that can be certified without arithmetic, and it isolates the cases where cancellation is doing real work. It also has a diagnostic use — a subvalue that vanishes for reason (iii) at one evaluation point and for reason (i) at another is a signal that the two points are being compared incorrectly, and this caught two errors during development.

The gauge. The totals at distinct normal forms are not equal, but they are constrained, and the constraint turns out to be representation-theoretic. Write $\mathcal{B}$ for the space of bidegree- $(2, 2)$ simultaneous-conjugation invariants of a pair of 3×3 matrices. Inside $\mathcal{B}$ put

$$u_1 = \text{tr}(A^2) \text{tr}(B^2) - \text{tr}(AB)^2,$$

$$u_2 = \text{tr}(A^2 B^2) - \text{tr}((AB)^2),$$

$$D = (\text{tr } A \text{ tr } B - \text{tr}(AB))^2 - ((\text{tr } A)^2 - \text{tr } A^2)((\text{tr } B)^2 - \text{tr } B^2),$$

and set $v := u_1 - 2u_2$ and

$$\Psi := 2v - D = 2u_1 - 4u_2 - D. \tag{4}$$

The third generator is where the obvious guess fails: $(\operatorname{tr} A \operatorname{tr} B - \operatorname{tr}(AB))^2$ by itself does not vanish on proportional pencils, and the discriminant completion D is the correct element. On the slot coordinates (u_1, u_2, D) the sampled points read $C = (-1, -1, 1)$, $R = (1, 0, 1)$, $T_4 = (3, 0, 5)$, the rank-one point $P = (0, 0, 0)$, and the entire compression family $(2, 0, 4)$; so $\Psi = 1$ at C , R and T_4 , and $\Psi = 0$ at P and on all of compression.

A warning about names, since they were assigned at different times and do not follow one convention. The subscript of T_4 counts *transvections* — it is the four-transvection point $x_3+ = x_0$, $x_4+ = x_1$, $x_7+ = x_1$, $x_8+ = x_2$ — and $\Psi(T_4) = 1$. The subscripts of X_4 and X_{-3} record their *gauge values*, $\Psi = 4$ and $\Psi = -3$. T_4 and X_4 are different points with different gauge values; we keep the names because the repository and the value records use them.

Proposition 5.3 (Uniqueness). *$\dim \mathcal{B} = 9$: the ten trace words of bidegree $(2, 2)$ satisfy exactly one relation, the polarised Cayley–Hamilton identity in rank 3. Inside $\mathcal{B}$, slab-equivariance with character $\det^2$ — that is $\mathcal{D}_A f + 2 \operatorname{tr}(A) f = 0$, and likewise in B — cuts out a one-dimensional space, spanned by Ψ . In particular the coefficient -4 on u_2 in (4) is forced, not fitted.*

Two properties follow from the shape alone, with no computation. Any $\det^2$ -equivariant co-variant vanishes identically on proportional pencils, since (A, cA) is a degenerate GL_2 -transform of $(A, 0)$; this accounts for the exact zeros at P . And Ψ is shift-invariant, which is what makes the normal form well posed.

Theorem 5.4 (The gauge is the Aronhold invariant). *Let I_6 be the fundamental degree-6 invariant of $\mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$ under $\operatorname{SL}_3 \times \operatorname{SL}_3 \times \operatorname{SL}_3$ — the lowest of the three basic invariants, of degrees 6, 9 and 12, in Vinberg’s classification of this θ -representation [18, 2]. Then*

$$I_6(I, A, B) = -6 \Psi(A, B), \quad (5)$$

an identity of polynomials in all 18 pencil indeterminates. Equivalently $\Psi = -I_6^{\text{prim}}(I, A, B)$ with coefficient 1, where I_6^{prim} is the primitive integral normalisation.

This is not the statement that Ψ is one component of the restriction: the restriction has no other component. Each ε in the first slot needs three distinct slab indices, so every three-block uses I , A and B once each and the slab multidegree is forced to $(2, 2, 2)$. The content of I_6 is 6, which is the factor in (5). As an independent check on the identification, our I_6 has exactly 1152 monomials on the generic $3 \times 3 \times 3$ tensor, matching the Aronhold invariant as tabulated by Bremner and Hu [2].

Theorem 5.5 (Totals law). *For every balanced direction $N = e_1 \otimes A + e_2 \otimes B$,*

$$\text{TOTAL}(N) = \Psi(N) \cdot 1\,152\,144\,000.$$

The measured totals are 1152144000 at C and at R , where $\Psi = 1$; 4608576000 at X_4 , where $\Psi = 4$; and 0 at the rank-one point P and at each of four compression points, where $\Psi = 0$. The law was stated as an empirical identity in an earlier version of this paper, tested at $\Psi = 1, 4, 0$ and at no other value; the proof below closes it. The one value it predicts that has never been measured is at X_{-3} , where $\Psi = -3$, and we record the prediction $\text{TOTAL}(X_{-3}) = -3\,456\,432\,000$ as a standing test of the two inputs the proof consumes.

For orientation: the functional is a GL_9 -Borel highest weight vector in $\operatorname{Sym}^{20} \operatorname{Sym}^3 \mathbb{C}^9$ — degree 20, not 18, because it lives at the $k = 1$ rung $\delta = 2 + 18$ of the conductor ray of Theorem 4.16, not in the degree of the invariant.

Proof of the totals law. Three facts go in, and the third is Proposition 5.3. The first two are recorded here.

Lemma 5.6 (Inputs). (a) *Content forces exactly four substituted legs, distributed as $t = (2, 2)$, so TOTAL is a polynomial of bidegree $(2, 2)$ on the 18-dimensional space of balanced directions $N = e_1 \otimes A + e_2 \otimes B$.* (b) *Let $Q \subset \mathrm{GL}_9$ be the stabiliser of the highest-weight line of h , a parabolic with Levi $\mathrm{GL}_3 \times \mathrm{GL}_6$ and $Q = \mathrm{Stab}(W)$ for the 6-dimensional coordinate subspace W , and let χ be its character on that line, $\chi(q) = \det(q|_{V/W})^8 \det(q|_W)^6$. If $u_{N'} = q u_N s$ with $q \in Q$ and $s \in H$ then $\mathrm{TOTAL}(N') = \chi(q) \mathrm{TOTAL}(N)$. Moreover $\Gamma_N := u_N^{-1}(W)$ satisfies $\Gamma_{N'} = s^{-1} \Gamma_N$, so the double coset of u_N is exactly the H -orbit of the net $\Gamma_N^\perp$ as a subspace of M_3 ; in the normal form of this section $\Gamma_N^\perp = \{[v; Av; Bv] : v \in \mathbb{C}^3\}$, whose tensor slabs are (I, A, B) .*

What was missing was the identification of χ , and it has two halves.

Lemma 5.7 (The character is a square). *Write $s = t^{-1}$, so that $q = u_{N'} t u_N^{-1}$ and $\Gamma_{N'} = t \Gamma_N$. Then*

$$\chi(q) = \det(q|_{V/W})^8 \det(q|_W)^6 = \det(q)^6 \det(q|_{V/W})^2 = \det(q|_{V/W})^2.$$

Proof. The first equality is $8 = 6 + 2$ and $\det q = \det(q|_{V/W}) \det(q|_W)$. For the second, u_N and $u_{N'}$ are unipotent, so $\det q = \det t$; and $\det t = \pm 1$ on H , since $X \mapsto aXb$ has determinant $(\det a \det b)^3 = 1$ and the transpose has determinant $(-1)^3 = -1$ on M_3 . Hence $\det(q)^6 = 1$ on both cosets of H . $\square$

The transpose coset therefore cannot contribute to χ — the same parity that closes the ray in Theorem 4.16, where $\det \mathfrak{g}$ is -1 on the transpose coset and the exponent is again even.

Lemma 5.8 (Slot dictionary). *$(V/\Gamma_N)^* = \Gamma_N^\perp$, and the basis of $\Gamma_N^\perp$ transported from $W^\perp$ by u_N is exactly the canonical parameter v of the normal form $\Gamma_N^\perp = \{[v; Av; Bv]\}$. Consequently $\det(q|_{V/W}) = \det(G)^{-1}$, where G is the change of parameter induced by t — that is, the basis change in the third tensor slot, the one on which H does not act. Explicitly, with $\alpha = a^{-T}$ and $\beta = b^{-T}$,*

$$G^T = \beta^T(\alpha_{00}I + \alpha_{01}A + \alpha_{02}B), \quad A' = (\beta^T(\alpha_{10}I + \alpha_{11}A + \alpha_{12}B))(G^T)^{-1},$$

and likewise for B' . Hence $\chi = \det(G)^{-2}$.

The two lemmas together are the identification $\chi \leftrightarrow \det^2$: the character is $\det^2$ in the net's parameter slot. Since I_6 is a semi-invariant of character $\det^2$ in each slot and the two M_3 -side factors contribute $(\det a \det b)^{-2} = 1$, the gauge obeys the same law,

$$\Psi(A', B') = \det(G)^{-2} \Psi(A, B) = \chi \cdot \Psi(A, B),$$

on both cosets of H — on the transpose coset because the transpose acts on the net by $Y \mapsto Y^T$, which swaps the first and third tensor slots, and S_3 acts on the one-dimensional degree-6 invariant space through the *trivial* character. (Had it been the sign character, Ψ would be anti-invariant where TOTAL is invariant and the law would be false. It is not: Ψ does not vanish on the nine-parameter family of tensors fixed by that swap.)

Proof of Theorem 5.5. Specialise t . Taking $\alpha = I$ and $\beta \in \mathrm{SL}_3$ gives $G = \beta^T$ with $\det G = 1$, so $\chi = 1$ and TOTAL is a simultaneous-conjugation invariant. Taking $\beta = I$ and α the unipotent sending $\mathrm{slab}_0 \mapsto \mathrm{slab}_0 + t \mathrm{slab}_1$ gives $(A, B) \mapsto (A(I + tA)^{-1}, B(I + tA)^{-1})$ with $\chi = \det(I + tA)^{-2}$, whose infinitesimal form is $\mathcal{D}_A \mathrm{TOTAL} + 2 \mathrm{tr}(A) \mathrm{TOTAL} = 0$; the slab_2 mirror gives the same in B . Both elements lie in H , since α and β are unimodular. So TOTAL lies in $\mathcal{B}$ and is slab-equivariant with character $\det^2$; by Proposition 5.3 that space is one-dimensional and spanned by Ψ , whence $\mathrm{TOTAL} = c \Psi$. Evaluating at C , where $\Psi = 1$ and the measured total is 1 152 144 000, gives c . $\square$

Two remarks on what the proof explains. First, C and R lie in *different* double cosets, so no symmetry of the setup carries one to the other; their equal totals looked like a coincidence and are now forced, because the group is too small to connect the points but the space of functions it constrains is one-dimensional. Second, the vanishing of the totals on the compression family is the classical degeneracy of the Aronhold invariant rather than a feature of the evaluation.

Remark 5.9 (Why $\det^2$). Nothing above is special to Ψ . For a deficit weight $\lambda' = (p, p, p, q^6)$ the same computation gives $\chi = \det(t)^q \det(q|_{V/W})^{p-q}$, so the character in the parameter slot is $\det^{p-q}$ and the transpose coset is harmless precisely when q is even. Here $p - q = 8 - 6 = 2$, which is why the governing object is the degree-6 Aronhold invariant. Section 6 works out what the whole family looks like.

Remark 5.10 (An arithmetic signature). Every value the functional has returned, at every point and every σ , is an integer multiple of $151\,200 = 2^5 3^3 5^2 7$. The reader can check this on the one table we give in full. At X_4 the eight orbit representatives carry weights $(4, 4, 8, 4, 8, 4, 2, 2)$, summing to the 36 subproblems, and cofactors

$$-2038, 4338, 3843, -258, 3567, -4372, -5188, -4552,$$

whose weighted sum is $30\,480 = \text{TOTAL}(X_4)/151\,200$. At C , R , Q and T_4 the corresponding leading cofactor is 719, and at X_{-3} it is 5907. The prime 719 organises the whole table and is what makes the refutation in Remark 5.11(a) sharp: it divides no other cofactor, so a proportionality $V_\sigma = W(\sigma)\Psi$ would force a non-integer where the engine returns an integer.

The common factor is not imposed anywhere in the computation, and we have no proof of it. Two further coincidences we flag without explaining: $\text{TOTAL}(C) = 2^7 3^4 5^3 \cdot 7 \cdot 127$ and $\text{TOTAL}(X_4) = 2^9 3^4 5^3 \cdot 7 \cdot 127$, so the prime 127 sits in the totals constant and nowhere else in the paper; and $\Phi_{18}(\det_3) = -2^{16} 3^7 5^3 7^2$ is 151 200 times $-5\,806\,080$, but it is a value in degree 18 and belongs to a different computation from the degree-20 table above, so we draw no inference from the shared factor.

Remark 5.11 (What fails). Three hypotheses about the functional were pre-registered and then refuted by their own tests. We record them because each was natural, each shaped the search, and each was killed by a single number. (a) The individual matrix elements V_σ^h do *not* inherit GL_2 -covariance: the engine returned $f1X_{400} = -308\,145\,600$ against a pre-registered $+434\,851\,200$, gate-confirmed by an independent run. The highest weight vector is a GL_9 -Borel object and the choice of scheme and of σ breaks the covariance. (b) The per- σ values are not simultaneous-conjugation invariants either. A parameter-free fit over $\mathcal{B}$ reached rank 9, the maximum on the feasible cone, and predicted $f1Y_{400} = +69\,854\,400$; the engine returned 0. The violated relation spreads over nearly every point, so it is not an outlier. (c) Both plane-cubic routes to the gauge are dead — there are no degree-2 invariants of ternary cubics, and three triangles realise $\Psi = 0, 1, 4$ — and the pencil cubic is insufficient, since Ψ requires the length-four word u_2 . None of this touches the results of Section 4: the conductor, the value at R , and the ray closure use no covariance assumption anywhere. What the refutations do is confine the phenomenon to the totals level, which is where Conjecture 5.5 states it.

6. OTHER WEIGHTS

The proof of Theorem 5.5 used the weight $\lambda' = (8, 8, 8, 6^6)$ only twice: in the counting lemma, and through the exponents 8 and 6 in χ . Both generalise, and the generalisation is not uniform — which is the interesting part. Throughout, $\lambda' = (p, p, p, q^6)$ and $m := p - q$.

Lemma 6.1 (Counting at a general weight). $|\lambda'| = 3p + 6q$, so $\delta = p + 2q$. Each copy of $\det_3$ contributes one leg from each matrix row, so the content before substitution is (δ, δ, δ) across

the three row blocks, against demands $3p, 3q, 3q$. Hence exactly $t_1 = t_2 = \delta - 3q = m$ legs are substituted out of each of rows 1 and 2, and TOTAL is a polynomial of bidegree (m, m) on the balanced cone.

Write $\mathcal{E}_m$ for the space of bidegree- (m, m) simultaneous-conjugation invariants of a pencil that are slab-equivariant with character $\det^m$, i.e. $\mathcal{D}_A f + m \operatorname{tr}(A)f = \mathcal{D}_B f + m \operatorname{tr}(B)f = 0$; this is the space in which Lemma 5.8 places TOTAL, and $\mathcal{E}_2 = \langle \Psi \rangle$ is Proposition 5.3. Let $R = \mathbb{C}[I_6, I_9, I_{12}]$ be Vinberg's invariant ring of $\mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$.

Theorem 6.2 (Restriction is an isomorphism). $I \mapsto I(I, A, B)$ is a linear isomorphism $R_{3m} \rightarrow \mathcal{E}_m$. Hence

$$\dim \mathcal{E}_m = \#\{(a, b, c) \in \mathbb{Z}_{\geq 0}^3 : 2a + 3b + 4c = m\},$$

which is 0, 1, 1, 2, 1, 3, 2 for $m = 1, \dots, 7$.

Proof. Each ε in the slab slot needs three distinct slab indices, so an invariant of degree $3m$ restricts in slab multidegree (m, m, m) : the map lands in $\mathcal{E}_m$. It is injective: if $I(I, A, B) \equiv 0$ then for any tensor with S_0 invertible the slot-2 action by S_0^{-1} gives $I(T) = \det(S_0)^m I(I, S_1 S_0^{-1}, S_2 S_0^{-1}) = 0$, and such T are dense. For surjectivity, given $f \in \mathcal{E}_m$ set $F(S_0, S_1, S_2) = \det(S_0)^m f(S_1 S_0^{-1}, S_2 S_0^{-1})$. Left multiplication of all slabs by g conjugates the normal form and contributes $\det(g)^m$; right multiplication by h fixes the normal form and contributes $\det(h)^m$; and a slab mixing α sends $S_0 \mapsto \tilde{P} S_0$ with $\tilde{P} = \alpha_{00} I + \alpha_{01} A + \alpha_{02} B$, so F is unchanged exactly because f is equivariant. So F is a semiinvariant of character $\det^m$ in each slot once it is polynomial. It is: F is regular off $\{\det S_0 = 0\}$, so its polar divisor lies there; that divisor is invariant, and the slab group moves $\{\det S_0 = 0\}$ to $\{\det(\sum c_k S_k) = 0\}$ for every c , so it lies in the locus where the slab pencil is identically singular, which has codimension at least 2. A divisor cannot, so there is no pole. $\square$

We verified the dimensions independently for $m \leq 7$: the ambient bidegree- (m, m) conjugation-invariant spaces have dimensions 2, 9, 25, 66, 149, 329 by an SL_3 character count, reproduced exactly as the rank of the trace-monomial evaluation matrix; the equivariant subspace was then computed as an exact nullspace over $\mathbb{Q}$ for $m \leq 4$, in $\mathbb{F}_p$ for $m = 5, 6$, and as the span of products of the generators for $m = 7$.

Corollary 6.3 (The uniqueness step, at every weight). $\dim \mathcal{E}_m = 1$ exactly for $m \in \{2, 3, 5\}$. So the argument of Theorem 5.5 determines TOTAL from a single measured total at $p-q = 3$ — where the governing object is the restriction of the degree-9 generator — and fails to determine it at $p-q = 4$, where $\mathcal{E}_4$ is spanned by Ψ^2 and the restriction of I_{12} and two measurements are needed. At $p-q = 1$, $\dim \mathcal{E}_1 = 0$: TOTAL vanishes identically on balanced directions, whatever q .

The degree-9 generator restricted to normal form, in primitive integral form, is a sum of thirteen trace monomials with coefficients $\pm 1, \pm 3, 6$; call it g_3 . It vanishes on compression nets, as every tensor invariant does — with compression correctly characterised as linear dependence of $\{I, A, B\}$ — and, more to the point, it vanishes at C, R, T_4, X_4, X_{-3} and P . Every point this programme has evaluated is therefore useless at $\lambda' = (9, 9, 9, 6^6)$, the $\delta = 3$ row, if the law holds there. Of the 262 144 balanced $\{0, 1\}$ directions, 87 660 have $g_3 \neq 0$; the smallest weight-feasible ones use four transvections, and we record one for the next evaluation:

$$G_{3a} = \{x_3+ = x_2, x_4+ = x_0, x_6+ = x_2, x_8+ = x_1\}, \quad g_3 = 3, \Psi = 0.$$

Odd q , and a prediction that is a zero. When q is odd, $\chi = (-1)^q \det(G)^{-m}$ picks up a sign on the transpose coset. The transpose acts on the net by $Y \mapsto Y^T$, which swaps the first and third tensor slots; the nets fixed by that swap are, in normal form,

$$A = \begin{pmatrix} 0 & 1 & 0 \\ p & r & s \\ u & v & w \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & 1 \\ u & v & w \\ y & z & c \end{pmatrix},$$

a nine-parameter family on which the transposed slab-zero is the identity, so $\det G = 1$ and $\chi = (-1)^q$. At odd q the double-coset theorem therefore forces TOTAL to vanish on the whole family — and since TOTAL lies in $\mathcal{E}_m$, this is a statement about $\mathcal{E}_m$ and not only about those directions.

Theorem 6.4 (Transpose vanishing). *g_3 vanishes identically on the self-transpose family; consequently a monomial $I_6^a I_9^b I_{12}^c$ restricts to zero there as soon as $b \geq 1$, and the rank of $\mathcal{E}_m$ on the family is 0 for m odd and $\lfloor m/4 \rfloor + 1$ for m even. Hence, at $\lambda' = (p, p, p, q^6)$ with q odd: if m is odd the transpose imposes nothing; if $m \in \{2, 4\}$ then $\text{TOTAL} \equiv 0$ on the entire balanced cone; and if $m \geq 6$ is even then TOTAL is confined to the subspace of elements divisible by I_9 — at $m = 6$, the line $\langle g_3^2 \rangle$.*

Equivalently: the degree-9 generator is anti-invariant under odd permutations of the three tensor slots, while I_6 and I_{12} are invariant.

Testing this needs a weight that is not empty. Most cheap odd- q weights are: computing $\text{amb}(\lambda', \delta) = \langle h_\delta[h_3], s_{\lambda'} \rangle$ gives 0 at $(3, 3, 3, 1^6)$, $(5, 5, 5, 1^6)$, $(4, 4, 4, 1^6)$, $(6, 6, 6, 1^6)$, $(4, 4, 4, 3^6)$, $(5, 5, 5, 3^6)$, $(6, 6, 6, 3^6)$ and $(7, 7, 7, 5^6)$. The cheapest odd- q weight with m even and a highest weight vector to evaluate is

$$\lambda' = (7, 7, 7, 3^6), \quad \delta = 13, \quad m = 4, \quad \text{amb} = 1,$$

and there Theorem 6.4 predicts $\text{TOTAL} = 0$ at *every* balanced direction. All five banked points C , R , T_4 , X_4 and X_{-3} are weight-feasible at this weight, so the prediction is not vacuous by infeasibility. We have not run it. It is the cheapest available test of the character computation on which this section rests, and it is independent of the value predicted at X_{-3} .

7. VERIFICATION PROTOCOL

Several of the results above are computational, and we regard the protocol as part of their content rather than as an appendix to it.

Exact arithmetic throughout. No floating point occurs anywhere in the pipeline. Multiplicities are computed by an exact streamed level algorithm over $\mathbb{Z}$: level states are packed into 54-bit masks held in machine words, the working set is an open-addressing table, and spilled runs are written to disk in a delta-varint code and merged. When the working set exceeds the memory budget the level is re-run in shards, with the shard count doubling until it fits; the answer is identical, only the schedule changes. Checkpoints are written atomically at every spill, so an interrupted run resumes at the last spill rather than at the last level.

Independent routes. The census values of Theorem 4.5 and the totals of Theorem 5.4 were each produced by two routes that share no code below the interface: the direct level algorithm, and a compressed route that quotients by the symmetry harvest first. The compression is not a heuristic — that the compressed sum reproduces the full sum is proved, by exhibiting the sign character on each orbit of subproblems and checking that the harvest is a group action with the claimed orbit sizes. The harvest reduces 36 subproblems to 18 and then to 11 inequivalent runs; the reduction is exact and the two routes agree on every value reported here.

Anchors. The pipeline is pinned at both ends by cases with independently known answers. At the small end, the same code path run on $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$ reproduces Cayley’s hyperdeterminant [5] on the standard test points: 0 at the two rank-one points, 0 at the W -state (rank 3, on the tangent locus), 1 at the GHZ point, and $9\,572\,836 = 3094^2 = (\det(u, u') \det(v, v') \det(w, w'))^2$ at a generic rank-two point — exactly, not to within rounding. This is an independent check on the invariant-theoretic layer, not merely on the arithmetic. At the classical end, the closed forms of Theorems 2.1 and 3.1 are reproduced by the general machinery. For Theorem 3.1 specifically, the torus criterion used in its proof and the multiplicity tables share no code: they agree on all 380 weights with $\delta \leq 10$ and on all 2361 weights with $m(\lambda) > 0$ and $\delta \leq 20$, on every member of the parity family up to $\delta = 28$, and on a 600-weight random sample at $\delta = 21, \dots, 28$. The corrected law was pre-registered, with named falsifiers, after the counterexample (17, 17, 2) and before any of those checks were run.

Canonical regression. A fixed regression suite is re-run before every reported value. Its published values are: the quadratic anchors $\text{quad} = 24$, $\text{quad}_0 = 0$, and $\text{quad}_q \text{ raw } 6 \times 4 = 24$; the $\det_3$ ladder $L_2 = 29/29/29$ through $L_6 = 1818118/2336283/2686868$; and, at the certification point, $L_7 = 54685987/100774838/141001840$ and $L_8 = 128027708/422952740/603408404$. Any deviation in any entry aborts the run. These numbers are stated here so that a reimplementa-tion can be checked against them without reproducing the whole computation.

Pre-registration. Predictions were committed to version control before the corresponding values existed, and the commit history is part of the record: for each of the results above one can check what was predicted, when, and whether it held. Two of the hypotheses so registered were refuted (Remark 5.11), and we consider their presence in the record to be evidence about the method rather than an embarrassment to it. A claim that was formed after seeing the value is marked as such wherever it appears.

8. OPEN PROBLEMS

Question 8.1 (Transport in general). Theorem 3.1 computes the conductor from boundary data alone, at every weight with $m(\lambda) > 0$ and with no degree bound, but not as the bare floor: $c = \lfloor \mu_{\max}/|w_N| \rfloor - \pi$ with π a parity defect supported on one congruence family. World A has $\pi \equiv 0$; World B does not. The natural general conjecture is therefore not that the floor is exact but that the correction is arithmetic in the same way: for an orbit closure with a smooth dense boundary orbit and a one-dimensional normal weight, is $c(\lambda) = \lfloor \mu_{\max}/|w_N| \rfloor - \pi(\lambda)$ with π determined by the order of the stabiliser character on the degenerating slot? In World B that order is 2 — symmetrisation over the S_3 in H acts as the pure parity filter $F(X, Y) + (-1)^{\lambda_1 - a_k} F(-X, -Y)$ — and π is exactly the obstruction to matching that parity against the residue $\mu_{\max} \bmod |w_N|$. Two concrete tests: does the same shape hold for $\sigma_r(v_d(\mathbb{P}^n))$ with a larger symmetric stabiliser, where the filter would be a character of order > 2 and π could exceed 1; and does the first deficit-positive weight of $\det_4$ (Question 8.5) carry a defect? The upper bound remains the block-order estimate and is general; what is not general is the assumption that Young symmetrisation always spares the top.

Question 8.2 (Determining the totals at $p - q \geq 4$). Corollary 6.3 says the equivariance argument determines TOTAL only when $\dim \mathcal{E}_m = 1$, i.e. at $p - q \in \{2, 3, 5\}$. At $p - q = 4$ it leaves the line through Ψ^2 and $I_{12}(I, A, B)$ undetermined, and two measured totals would fix it. Is there a second structural constraint that picks out the combination, or is the extra measurement genuinely needed? The natural guess — that the answer is a pure generator rather than a mixture, so that $\text{TOTAL} \propto I_{12}(I, A, B)$ with no Ψ^2 term — is testable with two

evaluations at $\lambda' = (10, 10, 10, 6^6)$ and would say something about which invariant the highest weight vector actually sees.

Question 8.3 (Degree 30, and the shape of the semigroup). Corollary 4.9 leaves exactly one degree open: is $30 \in \mathcal{E}(\det_3)$? Equivalently, is $S = \langle 18, 24 \rangle$ or $S = \langle 18, 24, 30 \rangle$? Unlike degree 24 this is not decided by one number: the ambient space $\mathbb{C}[\mathrm{Sym}^3 \mathbb{C}^9]_{30}^{\mathrm{SL}_9}$ is four-dimensional, so nonvanishing is still witnessed by a single evaluation but vanishing would require all four. Two remarks on what to expect. Every numerical semigroup with two generators is symmetric, so $S = \langle 18, 24 \rangle$ holds if and only if $\mathbb{C}[\overline{\Omega}]^{\mathrm{SL}_9}$ is Gorenstein, whereas $\langle 3, 4, 5 \rangle$ is not symmetric; we know no reason for Gorensteinness here and offer this only as the shape of the dichotomy. And by Remark 4.12 the answer names the last gap of $\frac{1}{6}S$, hence the exact conductor of $\mathbb{C}[\overline{\Omega}]^{\mathrm{SL}_9}$ in its normalisation $\mathbb{C}[\varphi]$: either 36 or 24.

Question 8.4 (The next conductor row). Theorem 4.16 settles $\delta = 2$. The row $\delta = 3$ requires either a theoretical reduction or a substantially larger computation; we believe the theory should come first, since the pattern of Section 3 suggests the answer is again a floor and the interesting content is the numerator.

Question 8.5 ($n = 4$ and beyond). The three known values are $e(\det_2) = 4$, $e(\det_3) = 18$ and $e(\det_4) = 16$. The even ones sit at n^2 and the odd one does not, which suggests the parity split that runs through [3] is the real division rather than any growth pattern in n . The first test is $n = 5$: is $e(\det_5) > 25$, and is it again $2n^2$? We expect the degree period to do as much work there as it did here — $5 \equiv 1 \pmod{4}$ gives $a = 1$ and $b = 5$, a coarser constraint than we enjoyed at $n = 3$, so the computation would be correspondingly harder. Closer to hand: what is the first deficit-positive weight of $\det_4$, and what is its conductor? A negative answer to Question 8.1 would most likely be visible there first.

DATA AVAILABILITY

All code, inputs, exact outputs, and the full commit history — including the pre-registration commits referred to in Section 7 — are available at

<https://github.com/swsethuraman/gct>

The regression values quoted in Section 7 are reproduced by the scripts in that repository without any of the long runs.

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